

Hypothetical one-year choice: JPY deposit rate 0%, USD deposit rate 5%, and $S_t = 150$ JPY per USD. Start with JPY 150,000.

	Keep yen	Convert to dollars
Deposit today	JPY 150,000	USD 1,000
Balance in one year	JPY 150,000	USD 1,050
Yen value if $S_T = 150$	JPY 150,000	JPY 157,500
Yen value if $S_T = 135$	JPY 150,000	JPY 141,750

Break-even exchange rate: $150,000/1,050 = 142.86$ JPY per USD.

A sufficiently strong yen erases the extra interest. An unhedged investor needs a view about the future exchange rate.

Source: Illustrative rates and payoffs. Ignore fees, taxes, and credit risk. S rises when the yen depreciates.

The foreign deposit pays $(1 + i_F)S_T/S_t$ in home currency. With risk-neutral investors and comparable assets, equilibrium equalizes *expected* returns:

$$1 + i_H = \frac{\mathbb{E}_t[S_T]}{S_t}(1 + i_F), \quad \mathbb{E}_t\left[\frac{S_T - S_t}{S_t}\right] \approx i_H - i_F.$$

- In the yen example, UIP requires $\mathbb{E}_t[S_T] = 150/1.05 = 142.86$.
- The higher-interest dollar must be expected to lose value against the yen enough to offset its interest advantage.
- $\mathbb{E}_t$ uses information available *today*. Tomorrow's realized exchange rate can be very different.

In the same yen example, frictionless CIP sets the one-year forward at 142.86 JPY per USD.

Object	What determines it?
Forward $F_{t,T}$	Today's spot and matched funding rates price an exchange contract.
UIP benchmark	Risk-neutral expected-return equality implies $\mathbb{E}_t[S_T] = 142.86$.
Investor forecast	A probability distribution based on the investor's economic assumptions and information.

An investor could reasonably forecast $S_T = 146$ while the forward is 142.86. The resulting expected gain involves exchange-rate risk.

A forecast needs a **currency pair, horizon, and economic story**. Compare developments in both economies.

Evidence	Economic question	Possible currency implication
Inflation, wages, jobs	Will the central bank keep rates higher than previously expected?	A higher expected relative policy path can support the currency today.
Growth, energy, trade	Is growth attracting investment, or raising the import bill?	Capital inflows and import payments might have opposite effects.
Risk and funding	Are investors seeking safety or closing leveraged positions?	Funding currencies can rise as borrowers repay them.

Observationally equivalent but counterfactually different: high interest rates have several causes, including tight monetary policy, high expected inflation, or compensation for risk. Each has different FX implications.

Source: Policy channel: Curcuro, Federal Reserve (2017). Funding channel: BIS Bulletin 90 (2024).

Hypothetical policy announcement: A central bank raises its rate by 25 basis points.

Before the decision	New information	Likely pressure, all else equal
No hike expected	More tightening than expected	Currency strengthens
A 50 bp hike expected	Less tightening than expected	Currency weakens
25 bp fully expected	Little news in today's rate	Future guidance

Where to look: Compare releases with survey forecasts. Compare the expected policy path before and after the news using rate futures or overnight indexed swaps (OIS).

Market prices contain risk premia

Source: Curcuro, Federal Reserve (2017). The scenarios are illustrative.

US home, UK foreign: S is USD per GBP. For this one-year illustration, hold the expected future spot rate at $\mathbb{E}_t[S_T] = 1.25$.

	Before the surprise	After the surprise
US / UK interest rate	3% / 3%	5% / 3%
Spot consistent with UIP	1.2500	$1.25(1.03)/1.05 = 1.2262$
Expected future spot	1.2500	1.2500

- **On the news:** S falls. The dollar appreciates immediately.
- **After that jump:** S is expected to rise from 1.2262 to 1.25. Expected dollar depreciation offsets its higher interest rate.

Return to the yen saver: $S_t = 150$ JPY per USD, one-year rates 0% in JPY and 5% in USD.

Hypothetical one-year scenario	Probability	S_T
US rates ease gradually, Japan normalizes	50%	145
US inflation persists, relative US rates stay high	30%	155
US slowdown and rapid closing of yen-funded trades	20%	135

- 1 Calculate $\mathbb{E}_t[S_T]$ and the expected yen return on the USD deposit.
- 2 How large is the yen loss in the last scenario?
- 3 Which new release would make you revise the probabilities?

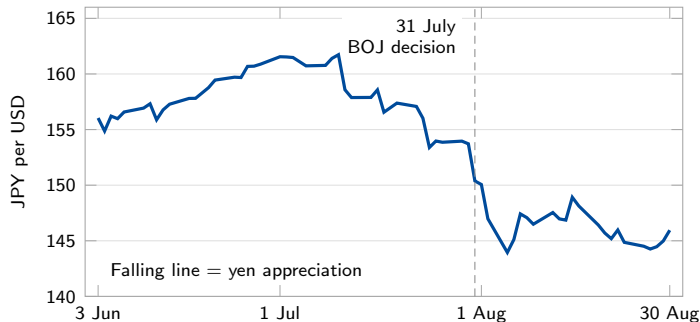
$$\mathbb{E}_t[S_T] = 0.5(145) + 0.3(155) + 0.2(135) = 146.$$

$$\mathbb{E}_t[R_{\text{USD deposit in JPY}}] = 1.05 \frac{146}{150} - 1 = \boxed{2.2\%}.$$

The JPY deposit earns 0%. The forecast implies a positive expected excess return from taking dollar risk.

$$\text{Return if } S_T = 135 : \quad 1.05 \frac{135}{150} - 1 = \boxed{-5.5\%}.$$

- The mean forecast differs from the UIP benchmark of 142.86.
- Weak US employment data might increase the probability of the last scenario. That lowers the mean forecast and the expected dollar return.
- A firm with a yen account payable may prefer certainty even when the expected dollar return is positive.



Background: Low Japanese rates made yen borrowing attractive for financing positions abroad.

Changing outlook: The BOJ raised its policy rate to around 0.25% on 31 July. Weak US data increased concern about a slowdown.

Amplification: Investors closing leveraged trades bought yen to repay funding. Yen appreciation magnified losses on remaining positions.

Why can a few days of exchange-rate losses overwhelm months of interest income?

Source: Federal Reserve H.10 via FRED, DEXJPUS, daily June–August 2024. Interpretation: BIS Bulletin 90 (2024).

The economic test: Does subsequent depreciation of high-interest currencies offset their interest advantage on average?

$$\underbrace{\frac{S_T - S_t}{S_t}}_{\text{realized home depreciation}} = \alpha + \beta \underbrace{\frac{F_{t,T} - S_t}{S_t}}_{\text{forward premium}} + \varepsilon_T.$$

- CIP, UIP, and rational expectations predict $\alpha = 0$, $\beta = 1$.
- Many historical short-horizon samples find $\beta < 0$. High-interest currencies often fail to depreciate enough, leaving positive average carry returns.
- Risk premia and systematic expectation errors are competing explanations. Estimates change across currencies, periods, and horizons.

Source: Bussière et al., The New Fama Puzzle.

Optional: calculate a Fama slope

All prices and wages use a common currency. Compare the same economy before and after a factory increases productivity.

	Before	After
Tradable output per worker-hour	1	2
Tradable price	10	10
Hourly wage in both sectors	10	20
Haircuts per worker-hour	1	1
Haircut price	10	20
Cost of one tradable + one haircut	20	30
Hours of work needed to buy that basket	2	1.5

The basket price $\uparrow$ 50%, but wages double. Higher prices can coexist with improved purchasing power.

Let A_T and A_N be output per worker-hour in tradables and nontradables. With a common domestic wage w and competitive labour-only production:

$$w = P_TA_T = P_NA_N, \quad \boxed{\frac{P_N}{P_T} = \frac{A_T}{A_N}}.$$

Productivity change	Relative service price P_N/P_T
Factory productivity doubles, salon unchanged	Doubles
Factory and salon productivity both double	Unchanged
Salon productivity doubles, factory unchanged	Halves

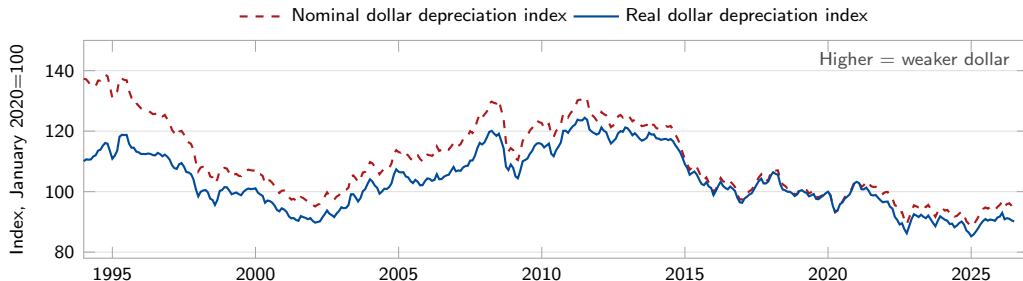
Local wages help explain expensive haircuts and meals. Rent, taxes, and product quality also matter, so a city price comparison is not a pure test of this model.

In the factory example, the home basket price rises from 20 to 30. Suppose the foreign basket price and nominal exchange rate stay fixed.

$$Q = \frac{SP_F}{P_H}, \quad \frac{Q_{\text{after}}}{Q_{\text{before}}} = \frac{20}{30} = \frac{2}{3}.$$

- Q falls: home real appreciation. Home services become more expensive relative to foreign services.
- This can be an equilibrium response to rising tradable productivity.
- With a nominal peg, relative prices adjust more. Otherwise, the nominal exchange rate can also adjust.

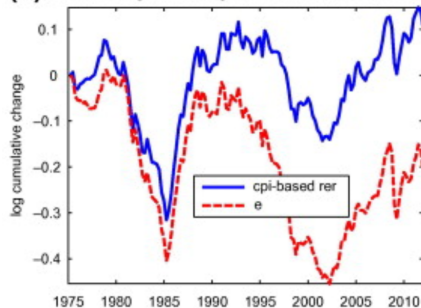
Updated evidence: Broad trade-weighted indices, January 1994–July 2026.



Reading the chart: Dollar swings appear in both series. Relative consumer prices create a slower-moving gap between them. Relative PPP does not hold continuously.

Source: BIS via FRED: NBUSBIS and RBUSBIS, downloaded 14 September 2026. Each series is inverted and rebased to January 2020=100 so that a rise means dollar depreciation, as with our Q . 100 is a reference date, not fair value.

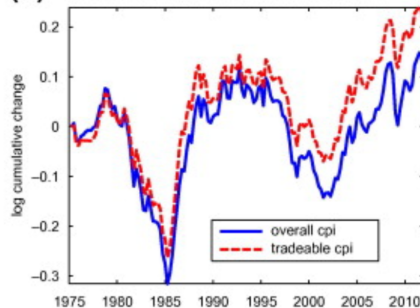
(a) US Trade Weighted Exchange Rate: Nominal and Real CPI-based



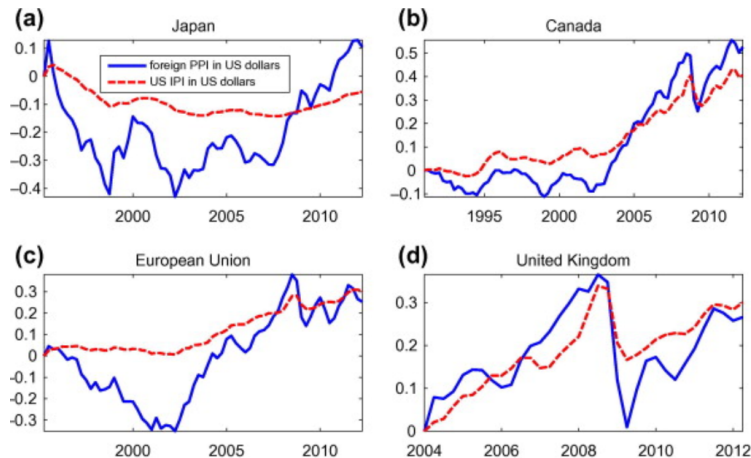
Left: Nominal and CPI-based real dollar movements share large swings. Consumer prices do not immediately offset currency movements.

Source: Burstein & Gopinath (2014), historical sample ending around 2012. Log cumulative changes, dollar depreciation upward.

(b) US CPI-based RER: Overall and tradeable-based



Right: Overall and tradable CPI-based real rates also move together. Nontradable services alone cannot explain all short-run real exchange-rate variation.



Blue: Foreign producer prices expressed in US dollars.

Red: US import prices from that source, also in dollars.

Firms can absorb currency changes in margins, and contracts can delay repricing. For example, dollar depreciation has incomplete pass-through to price paid by a US importer

Source: Burstein & Gopinath (2014). Historical log cumulative changes. PPI and import-price baskets differ.

Horizon	Useful starting point	What could change the forecast?
Days to weeks	News relative to expectations, positioning, funding conditions	A policy surprise or a rapid closing of crowded trades
Months	Relative policy paths, growth, inflation, and external payments	A different inflation or growth path than the one already priced
Years	Relative inflation and the sustainable real exchange rate	Productivity shifts, terms of trade, or persistent demand changes

A usable forecast states its assumptions and alternatives. Compare its performance with a simple unchanged-spot forecast, using only information available at the time.